CO2 ASSESSMENT Tool - 01

1) suppose that the data for analysis includes the attribute age the age values for the data toples are (in increasing order) 13, 15, 16, 16, 19, 20, 21, 22, 22, 25, 25, 23, 25, 30, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70

a) what is the mean of the data : what is the median!

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

13 + 15 + 16 + 16 + 19 + 20 + 20 + 21 + 22 + 25 + 25 + 25 + 25 + 30 + 33 + 33 + 35 + 35 + 35 + 35 + 36 + 40 + 45 + 46 + 52 + 70

$$= \frac{809}{27} = 29.9629$$

$$\text{Median} = \frac{n+1}{2} = \frac{27+1}{2} = \frac{\cancel{28}^{14}}{\cancel{2}} = 14 \text{ th values.}$$

14th values = 25 - median value

b) what is the mode of the data ? comment on the data's mudaility ?

25 & 35 are repeating for 4 time and it is binomel values.

datasets.

Similarity and Dissimilarity

proximity:

1) Data matrix

2) Dissimilarity

proximity measures:

$D(i,j) = \frac{p-m}{p}$ p= number of matche
m - number of variable

2) Given the following measurements for the variable age 18, 22, 25, 42, 28, 43, 33, 35, 56, 28 standardize the variable by the following:

compute the mean absolute deviation of age

$= \frac{330}{10} = 33$

| (x) Age | $\lvert x-\bar{x}\rvert$ |
|---|---|
| 18 | 15 |
| 22 | 11 |
| 25 | 8 |
| 42 | 9 |
| 28 | 8 |
| 43 | 10 |
| 33 | 0 |
| 38 | 8 |
| 56 | 23 |
| 28 | 5 |

$\Sigma \lvert x-\bar{x}\rvert$ $88/10 = 8.8$

3) Suppose that the data ~~for~~ analysis include the attribute age the age value for the data tople are 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 30, 33, 33, 35, 35, 35, 33 36, 40, 45, 46, 56, 70. Use smoothing by bin means to smoth the above data ~~to~~, using a bin depth of 3. illustrat your steps?

sol: Total No of values = 27

Tota No. of bines = $\frac{\text{total}}{\text{bin depth}}$ = 27/3 = 9.

| Bin | value | Bin mean | smooded v.o |
|---|---|---|---|
| $B_1$ | 13, 15, 16 | 44/3 = 14.67 | 14.67, 14.67. |
| $B_2$ | 16, 19, 20 | 55/3 = 18.33 | 18.33 |
| $B_3$ | 22, 20, 25 | 63/3 = 21 | 21 |
| $B_4$ | 25, 25, 30 | 72/3 = 24 | 24 |
| $B_5$ | 35, 35, 35 | 80/3 = 26.67 | 26.67 |
| $B_6$ | 35, 35, 35 | 101/3 = 33.67 | 33.67 |
| $B_7$ | 36, 40, 45 | 105/3 = 35 | 35 |
| $B_8$ | 36, 40, 45 | 121/3 = 40.33 | 40.33 |
| $B_9$ | 46, 52, 70 | 168/3 = 56 | 56 |

4) what ~~is the~~ standard Deviation

| $x$ | $x-500$ | $(x-500)^2$ |
|---|---|---|
| | | 90,000 |
| 200 | x-800 | 40,000 |
| 300 | -300 | 10,000 |
| 400 | -200 | 10,000 |
| 600 | -100 | 10,000 |
| 1000 | 500 | 250,000 |

The process of extracting useful patterns, knowledge,
... large datasets.

4) Use the two methods below to normalize the following group of data : 200, 300, 400, 600, 1000 (a) min-max normalization by setting min = 0 and max = 1 (b) z-score normalisation

A) min-max Normalization

$$x = \frac{x - min}{max - min} \cdot \frac{x - 200}{1000 - 200} = 800$$

| Original value | calculation | Normalized |
|---|---|---|
| 200 | 200-200 / 800 | 0 |
| 300 | 300-200 / 800 | 0.125 |
| 400 | 400-200 / 800 | 0.25 |
| 600 | 600-200 / 800 | 0.50 |
| 1000 | 1000-200 / 800 | 1.00 |

b) z-score Normalization

$$z = \frac{x - \bar{x}}{\sigma}$$

$$x = 200 + 300 + 400 + 600 + 1000 = \frac{2500}{5} = 500$$

2) c) what is the midrange of the data:

$$\frac{max + min}{2} = \frac{13+70}{2} = \frac{83}{2} = 41.5$$

d) can u find the first quartile (Q1) and third quartile (Q3) of the given for data?

Q = median value data

13 + 15 + 16 + 16 + 19 + 20 + 20 + 21 + 22 + 25 + 25 + 25 + 28

$$\frac{n+1}{2} = \frac{13+1}{2} = 14/2 = 7\text{th value}$$

Q3 = 30 + 30 + 30 + 35 + 35 + 38 + 36 + 40 + 48 + 4

the median of the upper 1s value is

Q = 35 the

the 7th value

Q3 − Q1 = 35 − 7 = 28

e) given the five number summary of the data?

min, Q1, median, Q3, max

13, 20, 25, 35, 70.